

FRE-GY 9743 Assignment 3 (Math Part)

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Q1. Normalized Forward Price

Bond: 9 years to maturity, coupon $c = 4\%$ semi-annual (coupon \$2 per period), spot $S = 100$, $r_{mkt} = 2\%$, forward horizon $T_d = 0.25$. Exactly 9 years to maturity means we are on a coupon date, so accrued interest is 0 and the dirty price is 100. The next coupon is in 6 months, after T_d , so no coupon is received during the carry period.

1) Forward price. Dirty forward $= S(1 + r_{mkt}T_d) = 100(1 + 0.02 \cdot 0.25) = 100.5$. At delivery, accrued interest is $2 \cdot \frac{3}{6} = 1$, so

$$BF = 100.5 - 1 = 99.5.$$

2) Conversion factor. Clean price of the 4% bond with 18 periods at a constant 6% yield (3% per period):

$$P = \sum_{k=1}^{18} \frac{2}{1.03^k} + \frac{100}{1.03^{18}} = 2 \cdot \frac{1 - 1.03^{-18}}{0.03} + \frac{100}{1.03^{18}} \approx 86.2465, \quad K = \frac{P}{100} \approx 0.86246.$$

3) Normalized forward.

$$\frac{BF}{K} = \frac{99.5}{0.86246} \approx 115.367.$$

4) Role of the conversion factor. Bonds in the basket have different coupons and maturities, so their forward prices are not comparable. K rescales each bond to the price it would have at a common 6% yield (the future's notional bond), so BF_x/K_x puts all bonds on the same footing as F . Since K is computed from the clean price on a coupon date, the forward price must also be clean, i.e. the accrued interest at delivery has to be subtracted from the dirty forward (as done above), otherwise bonds with different accrual at T_d are compared unfairly.

Q2. Cheapest-To-Deliver

1) Implied repo. With $F = 115.3670$, the implied clean forward is $F \cdot K = 115.3670 \times 0.86246 \approx 99.50$. Adding back accrued interest gives the dirty forward 100.50, and

$$r^{imp} = \frac{100.50/100 - 1}{0.25} = 2.00\%.$$

This equals r_{mkt} , so the bond in Q1 is exactly priced at the market repo (no arbitrage for this bond).

2) Choosing the CTD. Implied repo is the return from buying the bond, financing at repo, and delivering into the future. If $r_x^{imp} > r_{mkt}$, the bond can be bought and financed at r_{mkt} and delivered at a higher implied return, so it is cheap to deliver. The short chooses the bond with the *highest implied repo*, equivalently the lowest BF_x/K_x . Its implied repo is the one that matches the market future price, so $F = \min_x BF_x/K_x$.

Q3. Scenario Analysis

To predict the CTD N days ahead, take a scenario for the yield curve and the repo rate, reprice each bond at $t + N$ (spot, accrued, coupons received), recompute BF_x/K_x (or r_x^{imp}), and pick the bond with the lowest BF_x/K_x , equivalently the highest implied repo. The main driver is the 6% yield assumption in K :

- Yields above 6%: K overstates the value of high-duration bonds, so low-duration bonds (short maturity, high coupon) become CTD.
- Yields below 6%: the opposite, so high-duration bonds (long maturity, low coupon) become CTD.
- Repo specialness, curve slope and carry (coupon vs repo) shift the ranking too.

Why we care: the future price follows the CTD, so the futures hedge ratio and DV01 depend on which bond it is. The short also holds a delivery option (switching to another bond as the CTD changes) whose value must be included in pricing the future.